

# Cesar Rodriguez

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## EDUCATION

**University of Michigan** May 2023, US, MI Ann Arbor  
**Bachelor's of Science in Engineering in Computer Science - Honors Engineering - Magna Cum Laude** 3.55/4 GPA  
**Community Involvement:** Michigan Artificial Intelligence Safety Initiative, Society of Hispanic Professional Engineers  
**Relevant courses:**  
• Engineering: Machine Learning, Introduction to Artificial Intelligence, Web Systems, Human Centered Software, Introduction to Logic Design, Data Structures and Algorithms, Introduction to Computer Organization, Introduction to Computer Security  
• Philosophy: Mind and Machine, Descartes to Kant, Early Modern Philosophy, Critical Reasoning  
• Statistics: Statistics and Artificial Intelligence, Computational Methods, Introduction to Statistical Computing

## EXPERIENCE

**NextArena** US, IL, Chicago  
**Chief Executive Officer** July 2026 - Present

- Lead a team of four engineers developing a production-ready esports platform targeted for launch in 2026.
- Developed an automated outreach system to identify and contact content creators for promotional partnerships, scaling marketing efforts.

**Full Stack Developer - Technical Lead** January 2023 - July 2026  
• Implemented the company website in accordance with branding specifications, collaborating with designers to deliver an intuitive user experience.  
• Developed and documented frontend and backend REST APIs and database architecture to optimize data flow and improve performance.  
• Implemented a payment system supporting cryptocurrency and fiat transactions using row-level database locking for concurrency control, a state machine to enforce valid transaction transitions, and an append-only ledger for auditable transaction history.

**MIT Lincoln Laboratory** US, MA, Westford  
**Space Systems Analysis and Test Intern - Group 95** May 2023 - September 2023

- Developed a distributed PXE boot server using TCP, HTTP, and TFTP to automate Red Hat Enterprise Linux 8 deployments through Bash, Python, and C++.
- Reduced backend deploy times by 95% or more, and reduced setup complexity by 50% by testing on virtual machines and exporting to sensors.
- Implemented hardware-in-loop (HIL) testing automation strategy to analyze performance, reliability, extensibility, and long-term success.

**Intel Corporation** US, CA, Folsom  
**Software Engineering Intern - Cohort Lead** May 2022 - August 2022

- Created a plugin for an internal tool enabling integration of Intel's new security engine in Intel Firmware Images, improving firmware quality.
- Ensured compatibility with external firmware images by developing Python and Bash tooling that standardized image generation.
- Refactored C++ class hierarchies to improve maintainability and interoperability with the new firmware engine.
- Conducted code reviews and utilized pair programming to develop and implement new features for the new firmware engine.

**Intel Corporation** US, IL, Chicago  
**Software Engineering Intern - Cohort Lead** May 2021 - August 2021

- Standardized stress testing operating system for validating 14th generation central processor units, resulting in increased performance and reliability.
- Modernized legacy Python tooling across multiple teams, earning four Intel recognitions and reducing execution time by approximately 4%.
- Developed and evaluated simulated models to test CPU functionality on FPGA, resulting in a report of findings and recommendations to the team.
- Trained five international team members in Malaysia on usage and implementation of FPGA simulated model for testing CPU functionality.

## PROJECTS

**University of Illinois Chicago** US, IL, Chicago  
**Freelance Data Analyst** December 2017 - May 2022

- Extracted unstructured data from image sources to produce human-readable dataset for use by non-technical Museum Studies Program faculty.
- Cleaned, structured, and analyzed datasets supporting research into government accountability.
- Designed data structures to promote ease of use, scalability, and accessibility on a website accessed by tens of thousands of users.

**University of Michigan** US, MI, Ann Arbor  
**Research Assistant, Cardiovascular Technology** October 2019 - May 2020

- Interviewed healthcare professionals to identify workflow bottlenecks in emergency cardiovascular response and inform application design.
- Created an application to improve communication between medical personnel that reduced response time from 7 minutes to 5.5 minutes by automating note-taking and streamlining the communication process.

## SKILLS

**Languages:** Python, TypeScript, JavaScript, C++, C, SQL, Java, R, MATLAB, Verilog

**Frameworks:** React, Django

**Technologies & Tools:** Node.js, Git, Linux, Visual Studio, Cadence Perspec, Arduino, AutoCAD, Photoshop

**Operating Systems:** Windows, Ubuntu, Red Hat Enterprise Linux, macOS